

METAL SOLUTIONS

EOS Titanium TiCP Grade 2

Material Data Sheet

EOS TITANIUM TiCP GRADE 2

The parts built with EOS Titanium TiCP grade 2 powder have chemical composition corresponding to ASTM F67. The parts have good strength-to-weight ratio, corrosion resistance and ductility. Parts built with EOS Titanium TiCP grade 2 powder can be machined, shot-peened and polished in as-built and heat treated states. Due to the layer-wise building method, the parts have a certain anisotropy.

MAIN CHARACTERISTICS

- Good strength-to-weight ratio
- Good corrosion resistance & ductility

TYPICAL APPLICATIONS

- Parts for medical and other industries

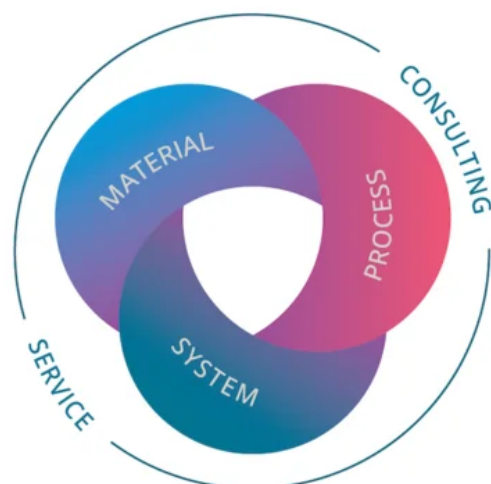
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards



POWDER PROPERTIES

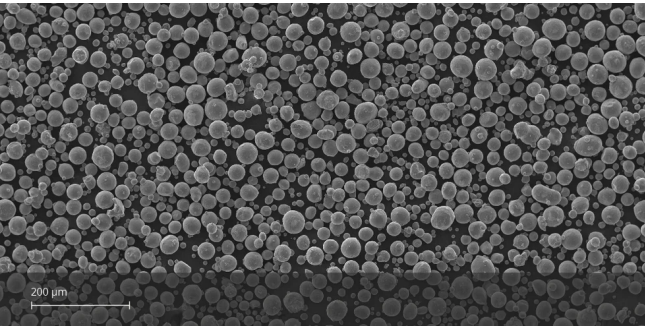
EOS Titanium TiCP Grade 2 powder corresponds to ASTM F67

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
N	0	0.03
C	0	0.08
H	0	0.015
Fe	0	0.3
O	0	0.25
Ti		Balance

Powder Particle Size

Generic Particle Size Distribution	38 - 45 μm
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HEAT TREATMENT

Description

Heat treatment procedure

Steps

Heat treatment in 700 °C ($\pm$ 10 °C) for 2 h ($\pm$ 0.5 h) under argon.

EOS Titanium TiCP for EOS M 290 | 30 µm

EOS M 290 - 30 µm

System Setup	EOS M 290
EOS Material set	TiCP 30µm FlexLine
Software Requirements	EOSPRINT 1.3 or newer
	EOSYSTEM 2.3 or newer
Recoater Blade	HSS (High Speed Steel)
Inert gas	Argon
Sieve	63 µm

Additional Information	
Layer Thickness	30 µm
Volume Rate	5.0 mm³/s

Microstructure of the Produced Parts

Density EN ISO 3369	Thickness	Result	Number of Samples
Average Density	30 µm	≥ 4.5 g/cm³	-

Mechanical Properties

Mechanical Properties As Manufactured

EN ISO 6892-1 Room Temperature 30 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	560	660	22	≥ 30	-	-
Horizontal	560	660	22	≥ 30	-	-

The numbers are average values and are determined from samples with horizontal and vertical orientation.

The numbers are average values and are determined from samples with horizontal and vertical orientation.

Mechanical Properties Heat Treated

EN ISO 6892-1 Room Temperature 30 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	445	570	26	≥ 30	-	-
Horizontal	445	570	26	≥ 30	-	-

The numbers are average values and are determined from samples with horizontal and vertical orientation.

Heat treatment in 700 °C (± 10 °C) for 2 h (± 0.5 h) under argon.

The numbers are average values and are determined from samples with horizontal and vertical orientation.

Hardness

EN ISO 6507 Heat Treated	
Value	195
Unit	HV5

Surface roughness after shot peening

Post-processed	
Value	Ra < 10 µm; Rz < 55 µm

HEADQUARTERS

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Status as of 18.08.2026. Subject to technical modifications. EOS is certified according to ISO 9001.

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